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Project S6: Final review

Clock synchronized by RF antenna

THE « DREAM » TEAM

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SUMMARY

PHASE 1 : Definition & Analisis

A – Forecast Schedule

B – Functional Analisis

C – Components Choice

D – Implementation schematics

PHASE 2: PCB Conception 2D/3D

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B – Feedback

C – Greetings

FORECAST SCHEDULE

Nom	Date de début	Date de fin
Phase 1 : Analyse & Définition	02/02/2026	09/02/2026
Analyse Fonctionnelle	02/02/2026	09/02/2026
Recherche et choix des composants	02/02/2026	09/02/2026
Phase 2 : Réalisation Technique	16/02/2026	13/04/2026
Commande matériel	16/02/2026	16/02/2026
Réception matériel	09/03/2026	09/03/2026
Conception PCB KiCAD (Alimentation, Acquisition, Affichage)	16/02/2026	30/03/2026
Revue de projet Mi-parcours	13/04/2026	13/04/2026
Phase 3 : Intégration & Clôture	04/05/2026	25/05/2026
Programmation composants STM32	04/05/2026	11/05/2026
Soudure des composants (après fabrication PCB pendant les vacance...	04/05/2026	11/05/2026
Assemblage final et test du mode économie d'énergie.	18/05/2026	18/05/2026
Finalisation du rapport + Soutenance.	25/05/2026	25/05/2026

3 phases: **Definition**, **Conception**, **Fabrication**

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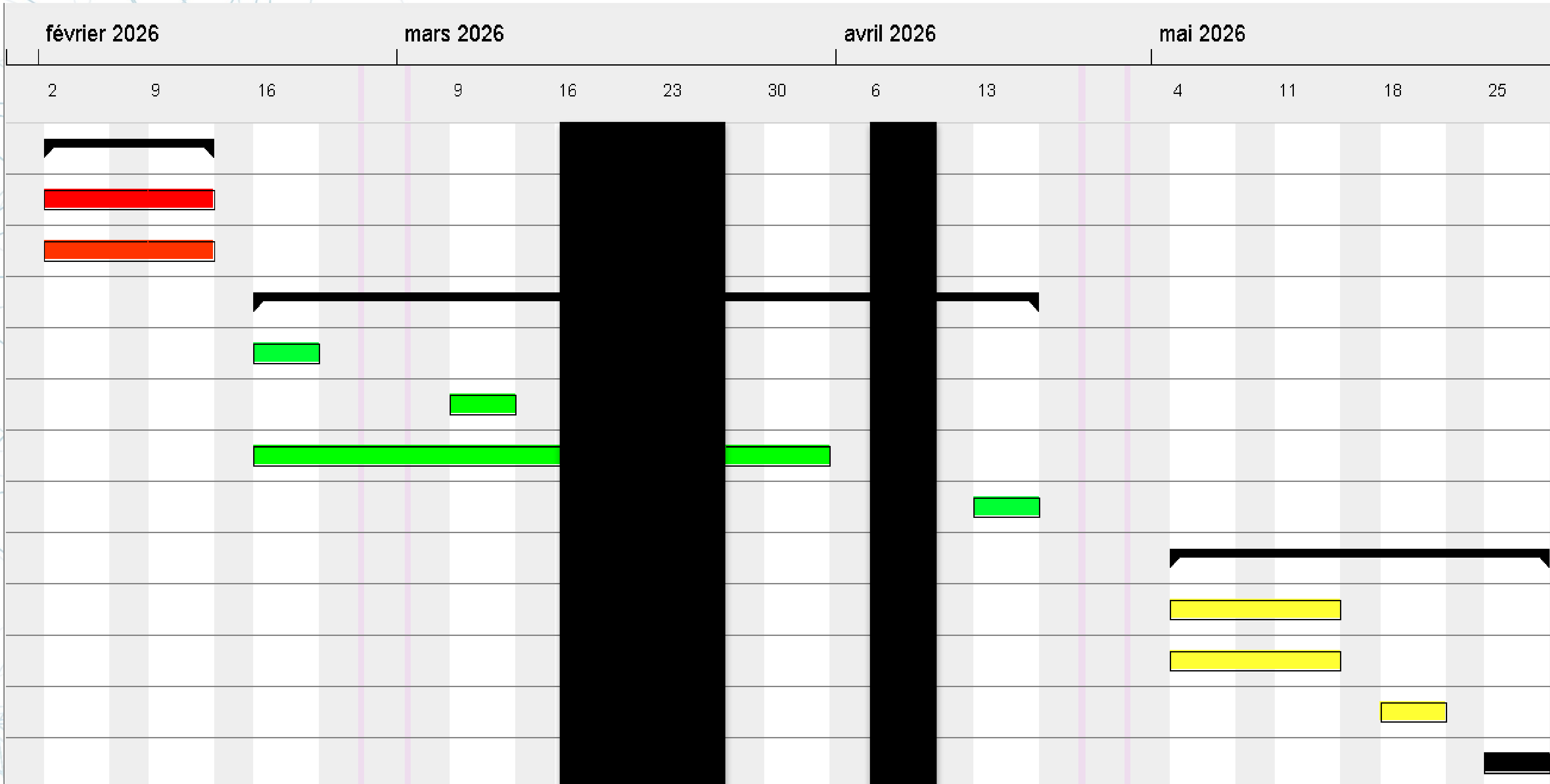
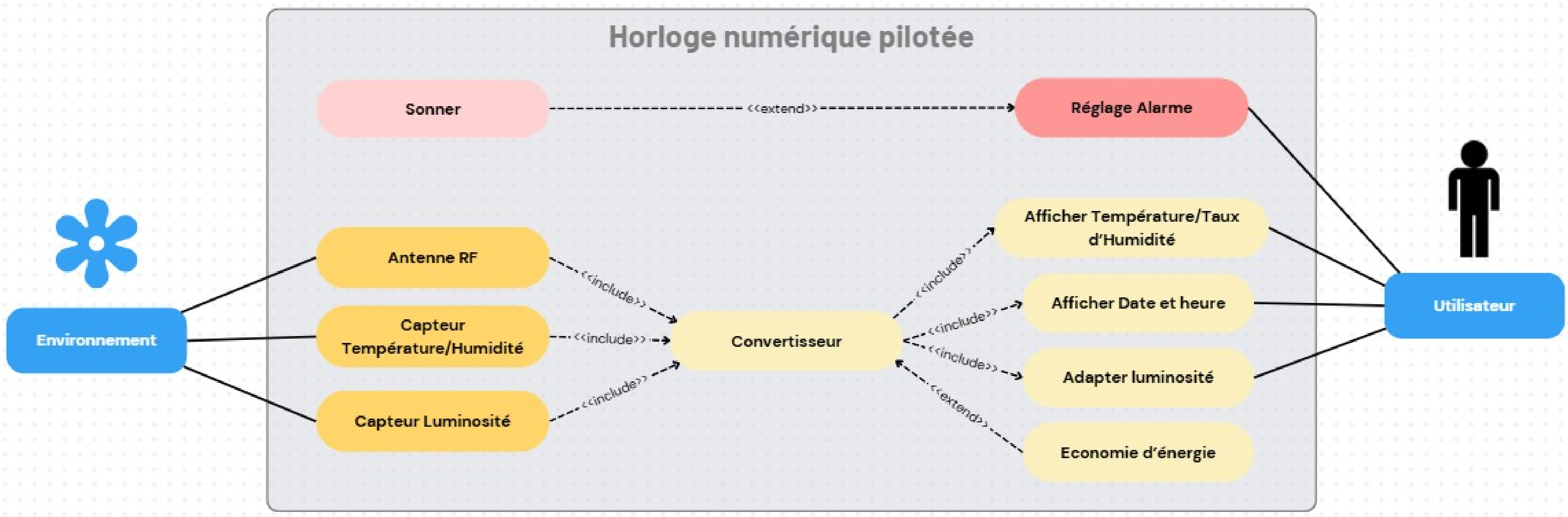
FORECAST SCHEDULE



<u>Start</u>: Monday 2nd february 2026

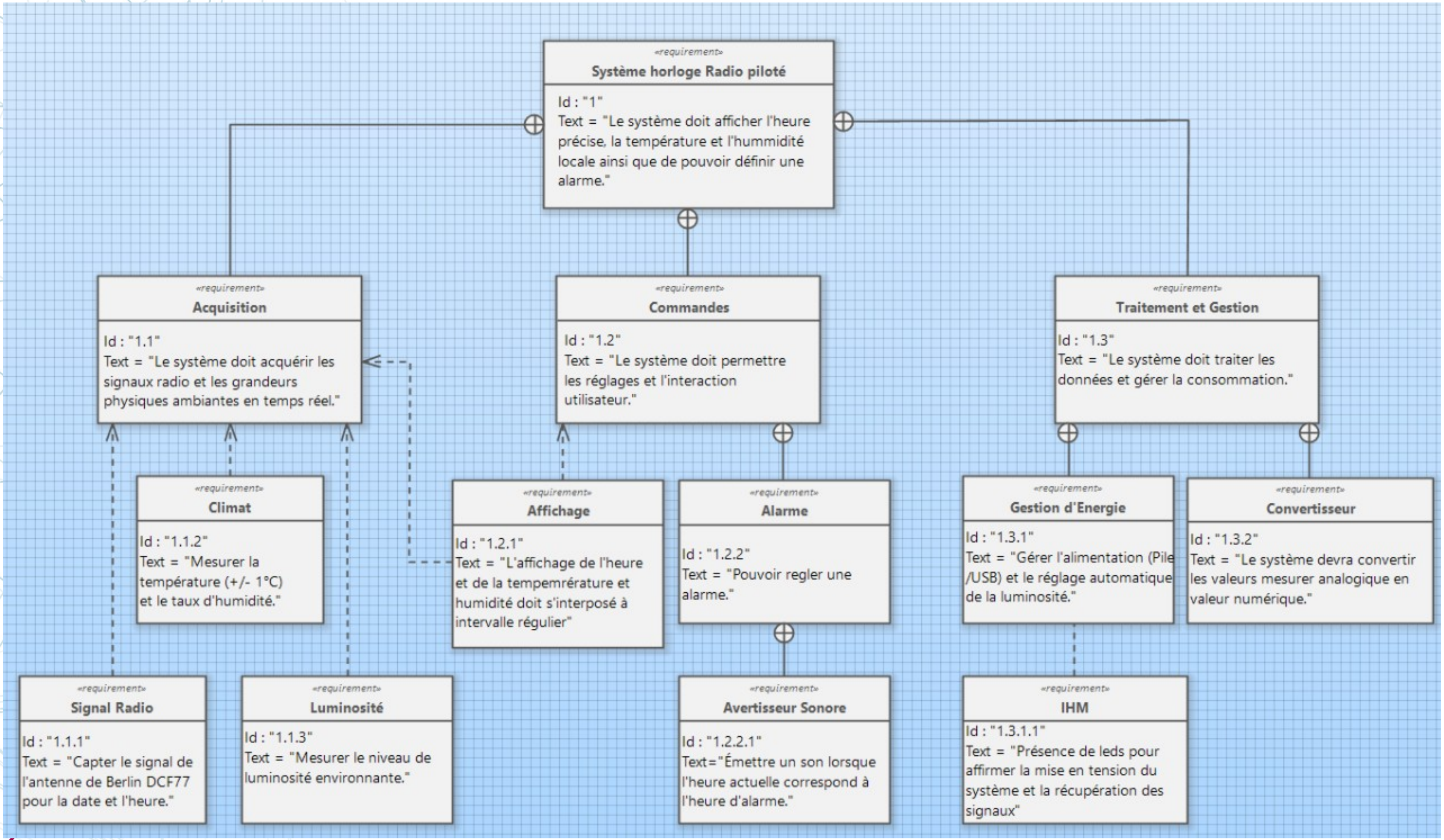
<u>End:</u> Thursday 29th may 2026

FUNCTIONAL ANALISIS



General structure of the project

FUNCTIONAL ANALISIS



Specifications simplified by a requirements diagram

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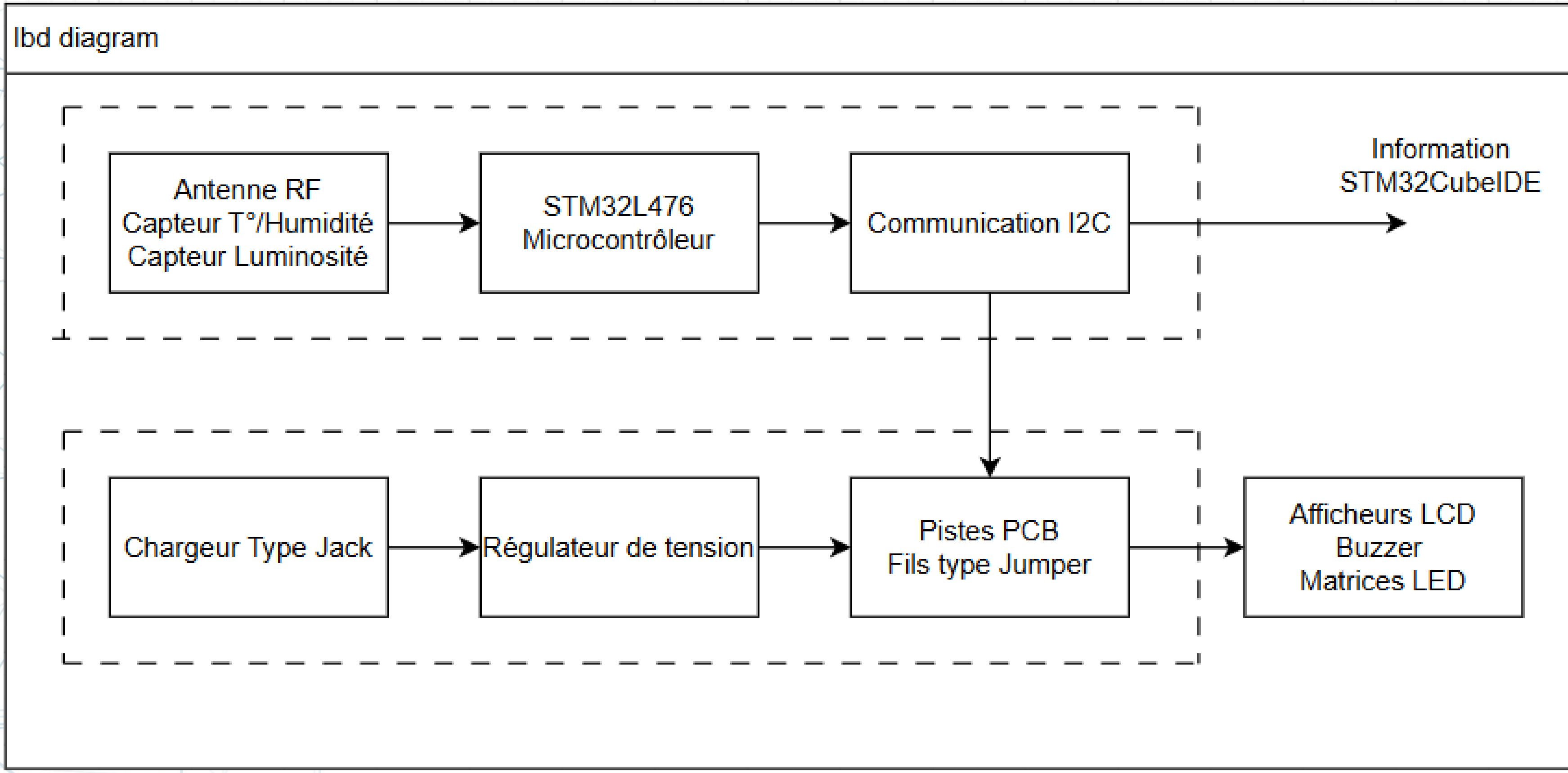
FUNCTIONAL ANALISIS



Flows of energy and data involved

COMPONENT CHOICE

Acquisition:
1x SHT41L (Humidity + Temperature sensor)
1xTEMT6000X1 (brightness)
1x DCF77 (RF Antenna)

1 ACQUISITION PCB

Power supply:
1xBU33D5 (5V-3,3V regulator)
1x Dissipateur (fot L78)
1x L78 (12V-5V regulator)
1x Pj-102 plug

1 POWER SUPPLY PCB

display:
1x Buzzer
1x I2C displayer
5x LED matrice

1 DISPLAY PCB

Main suppliers: RS Components and Farnell

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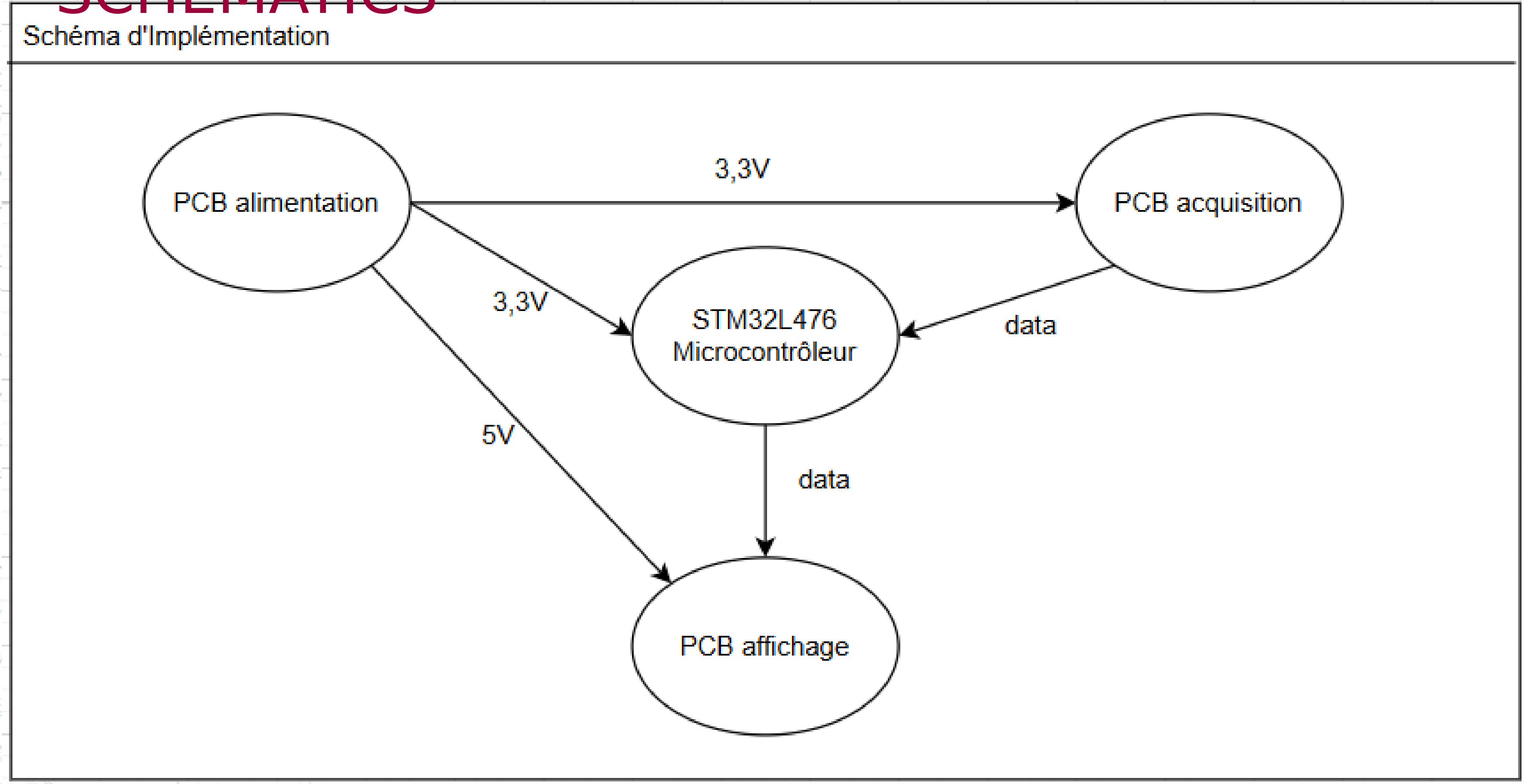
IMPLEMENTATION SCHEMATICS



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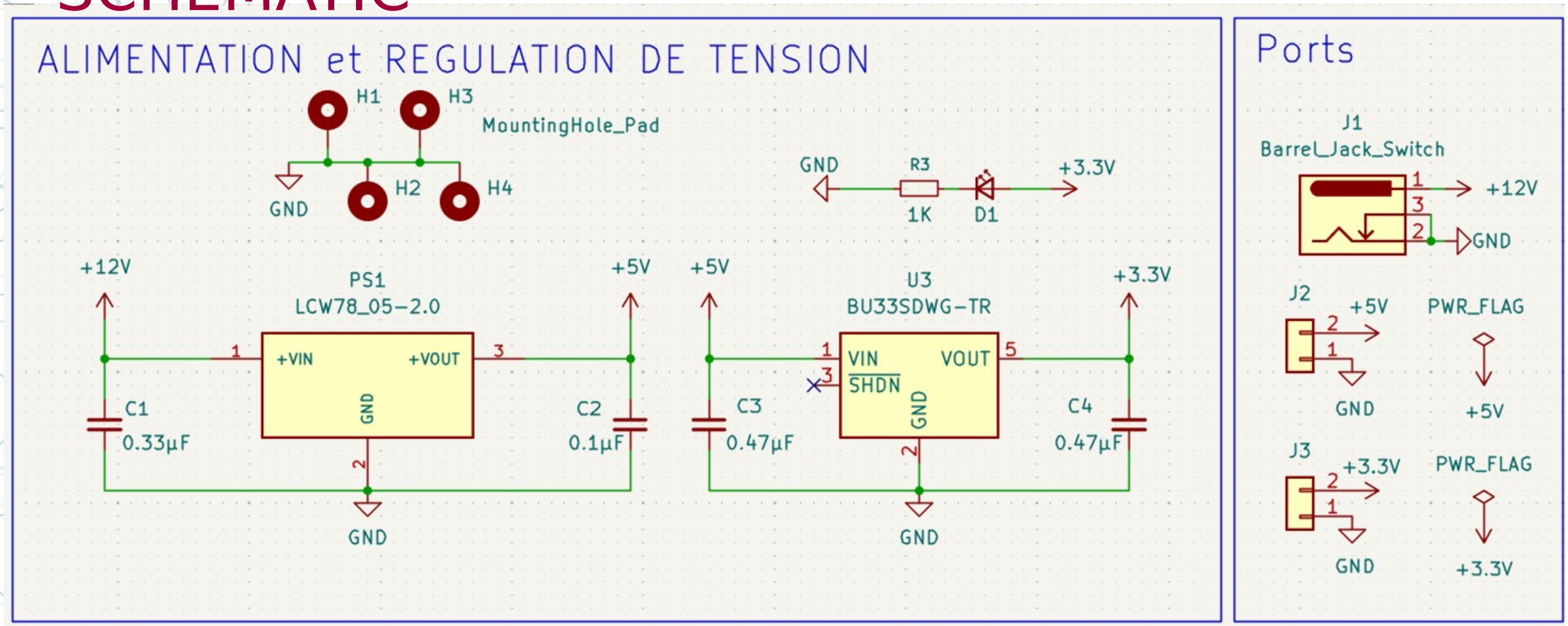
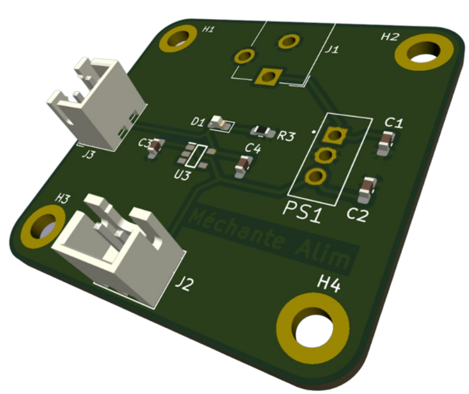
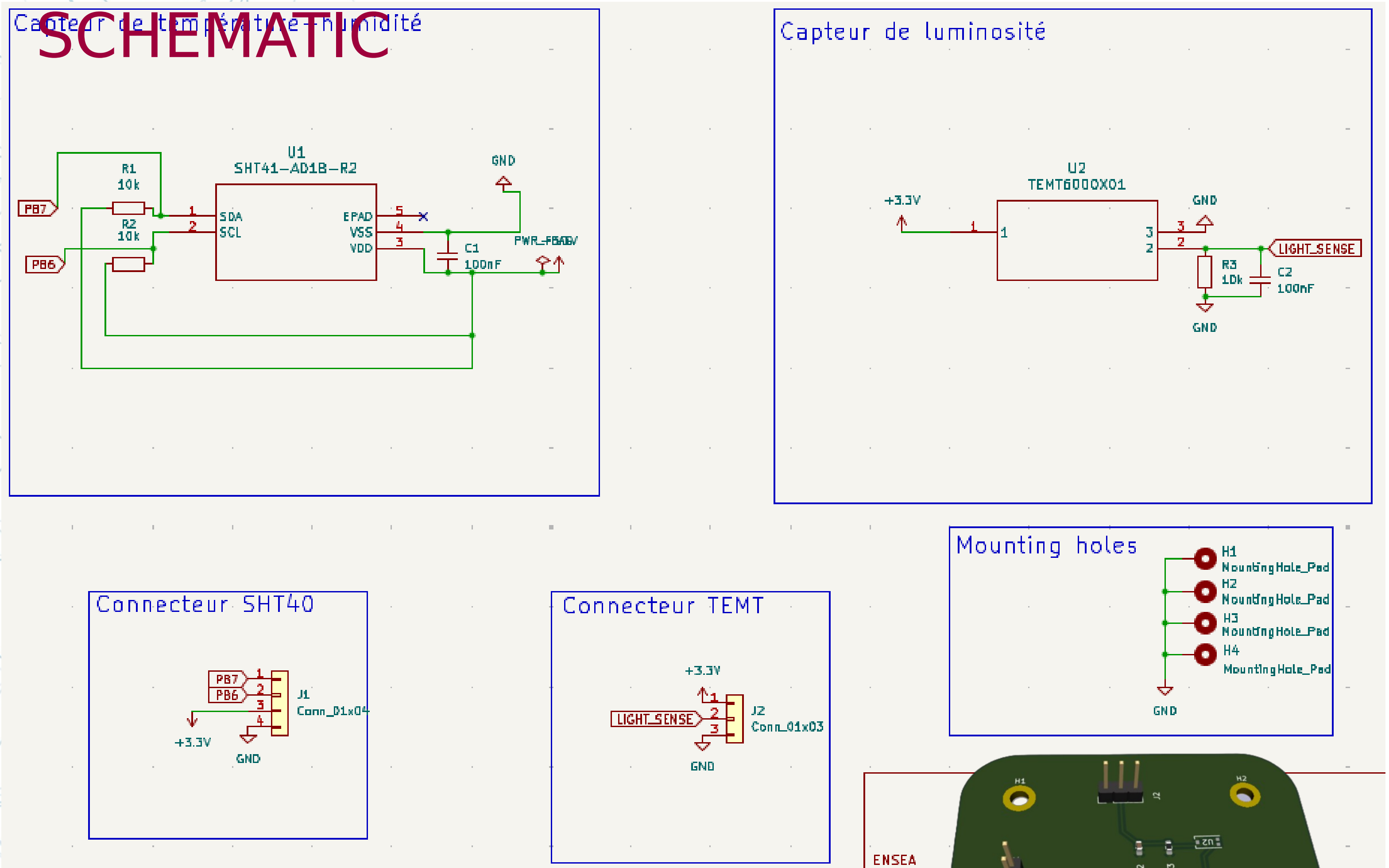
ALIMENTATION SCHEMATIC

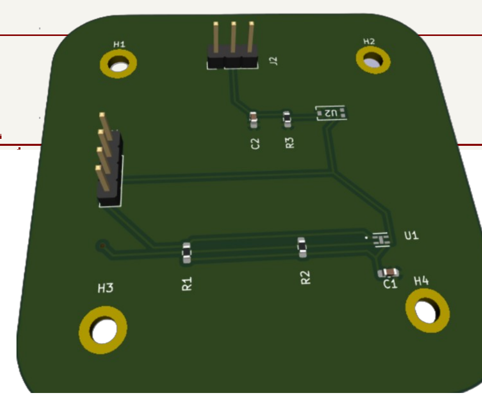


Nicknamed “Méchante Alim” by its creator



ACQUISITION SCHEMATIC





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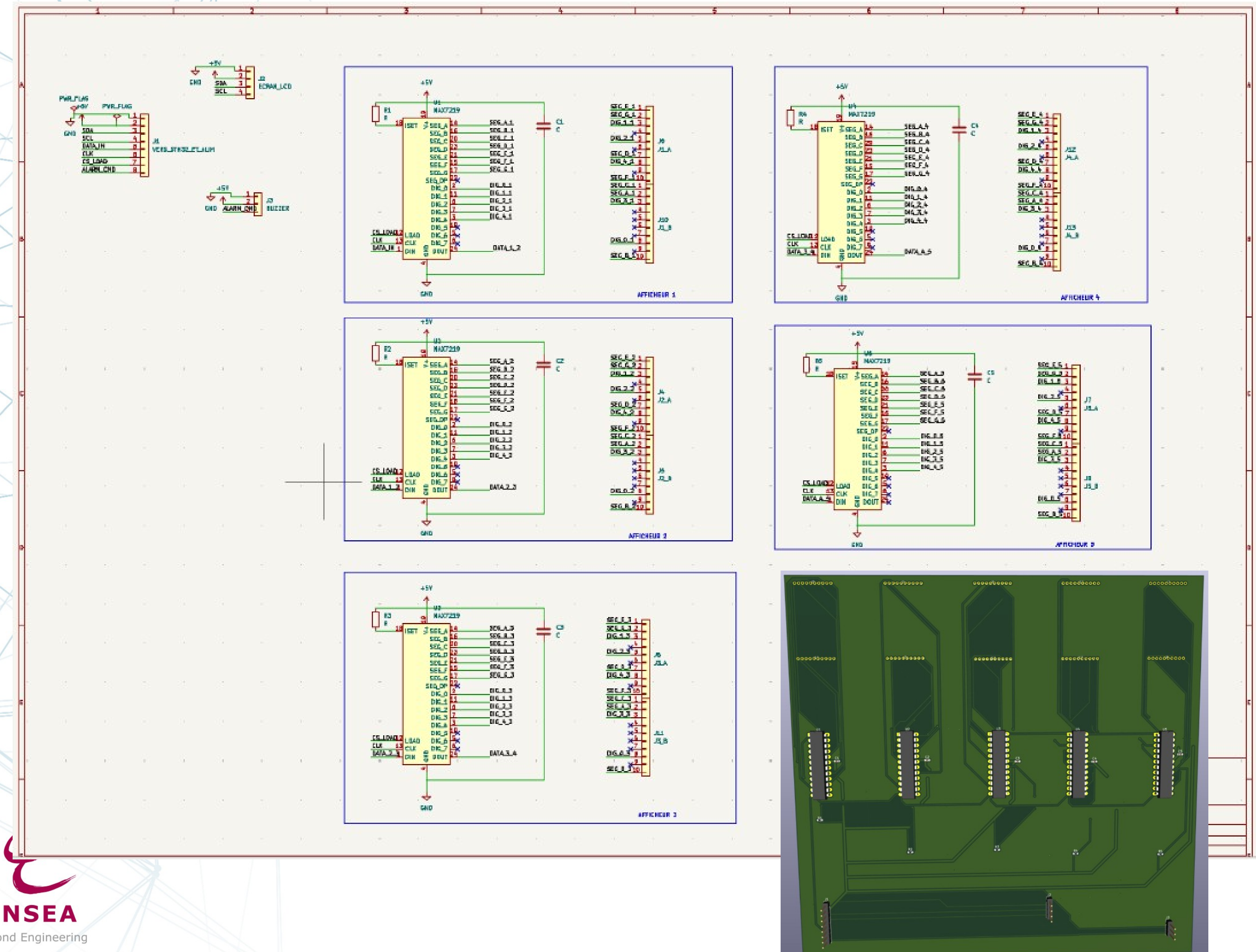
DISPLAY SCHEMATIC



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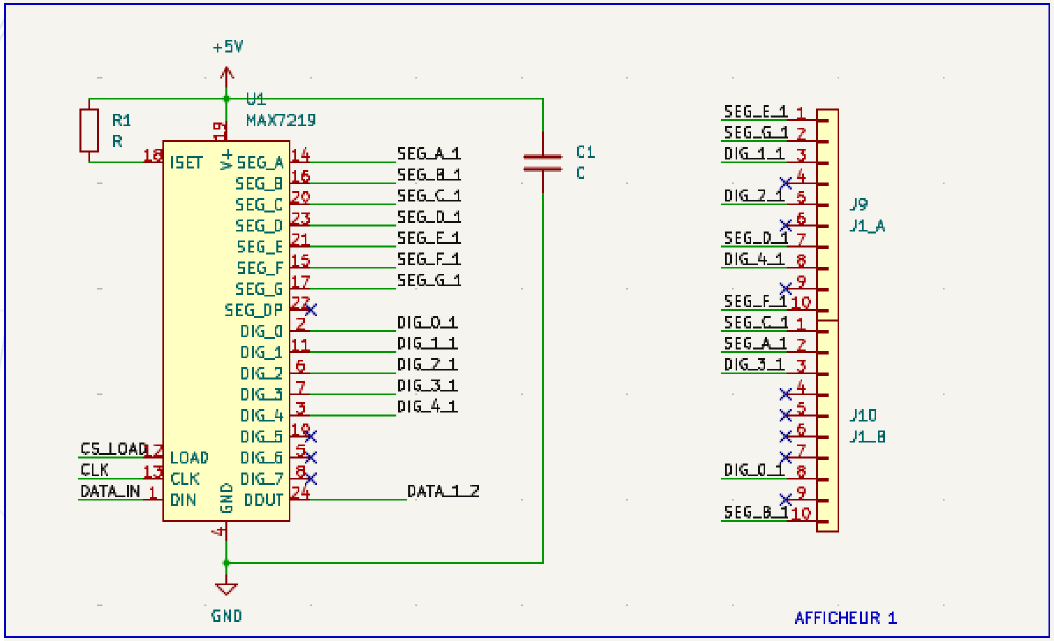
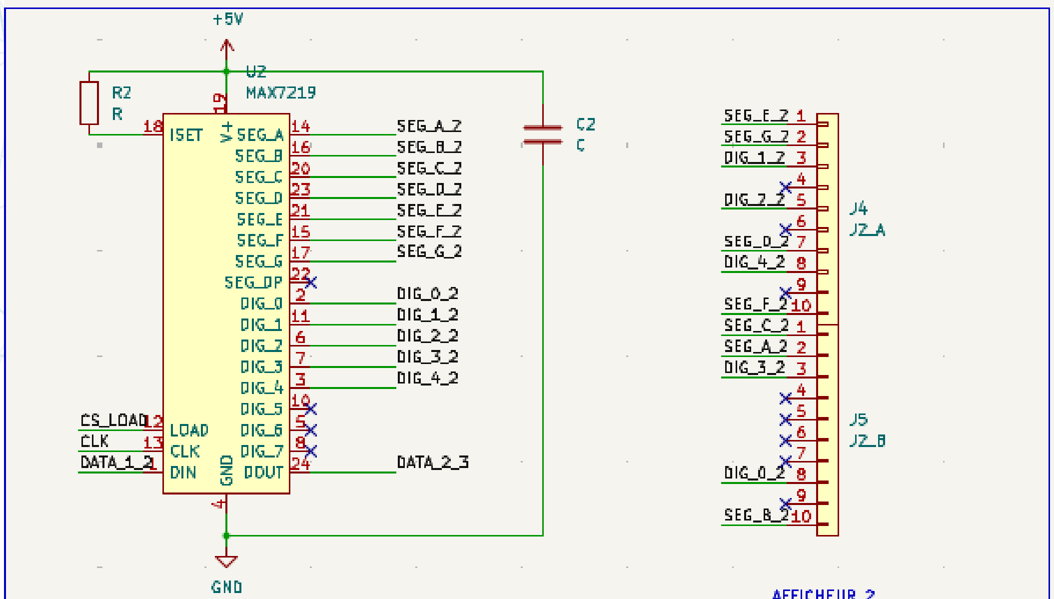

+5V
U1
MAX7219
R1
R
C1
C
ISET
V+
SEG_A
SEG_B
SEG_C
SEG_D
SEG_E
SEG_F
SEG_G
SEG_DP
DIG_0
DIG_1
DIG_2
DIG_3
DIG_4
DIG_5
DIG_6
DIG_7
LOAD
CLK
DIN
GND
DOUT
CS_LOAD
CLK
DATA_IN
SEG_A_1
SEG_B_1
SEG_C_1
SEG_D_1
SEG_E_1
SEG_F_1
SEG_G_1
DIG_0_1
DIG_1_1
DIG_2_1
DIG_3_1
DIG_4_1
DATA_1_2
GND
SEG_E_1
SEG_G_1
DIG_1_1
DIG_2_1
SEG_D_1
DIG_4_1
SEG_F_1
SEG_C_1
SEG_A_1
DIG_3_1
DIG_0_1
SEG_B_1
J9
J1_A
J10
J1_B
AFFICHEUR 1


+5V
U2
MAX7219
R2
R
C2
C
ISET
V+
SEG_A
SEG_B
SEG_C
SEG_D
SEG_E
SEG_F
SEG_G
SEG_DP
DIG_0
DIG_1
DIG_2
DIG_3
DIG_4
DIG_5
DIG_6
DIG_7
LOAD
CLK
DIN
GND
DOUT
CS_LOAD
CLK
DATA_1_2
SEG_A_2
SEG_B_2
SEG_C_2
SEG_D_2
SEG_E_2
SEG_F_2
SEG_G_2
DIG_0_2
DIG_1_2
DIG_2_2
DIG_3_2
DIG_4_2
DATA_2_3
GND
SEG_E_2
SEG_G_2
DIG_1_2
DIG_2_2
SEG_D_2
DIG_4_2
SEG_F_2
SEG_C_2
SEG_A_2
DIG_3_2
DIG_0_2
SEG_B_2
J4
J2_A
J5
J2_B
AFFICHEUR 2

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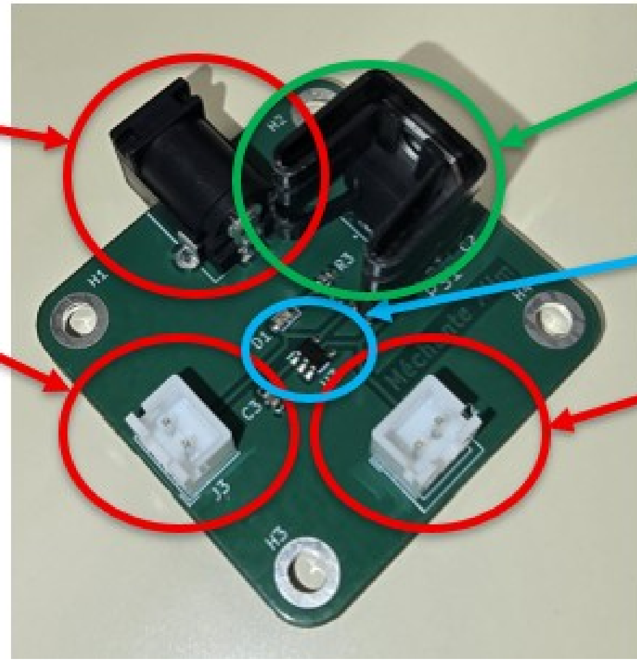
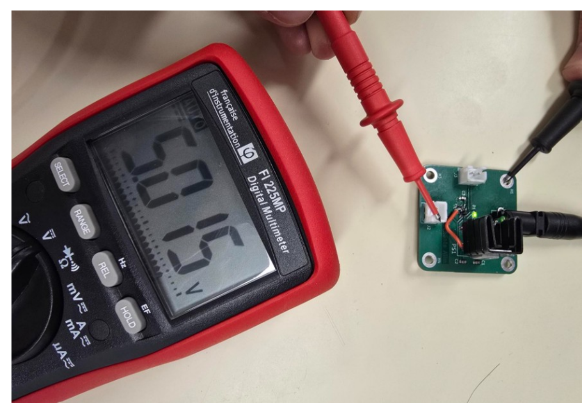
ALIMENTATION



gap justified by tolerances of the components

The PCB is functional for our application



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ACQUISITION

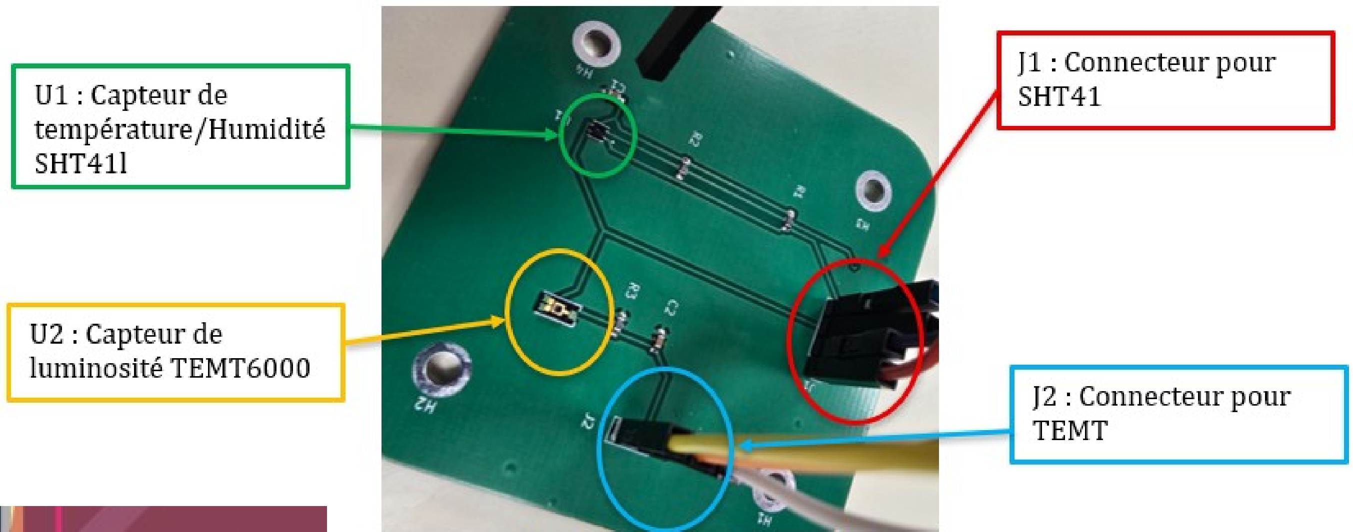
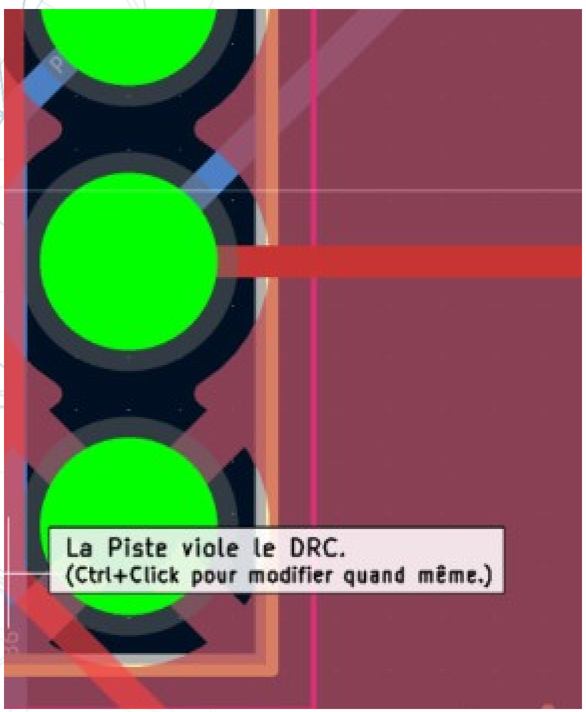


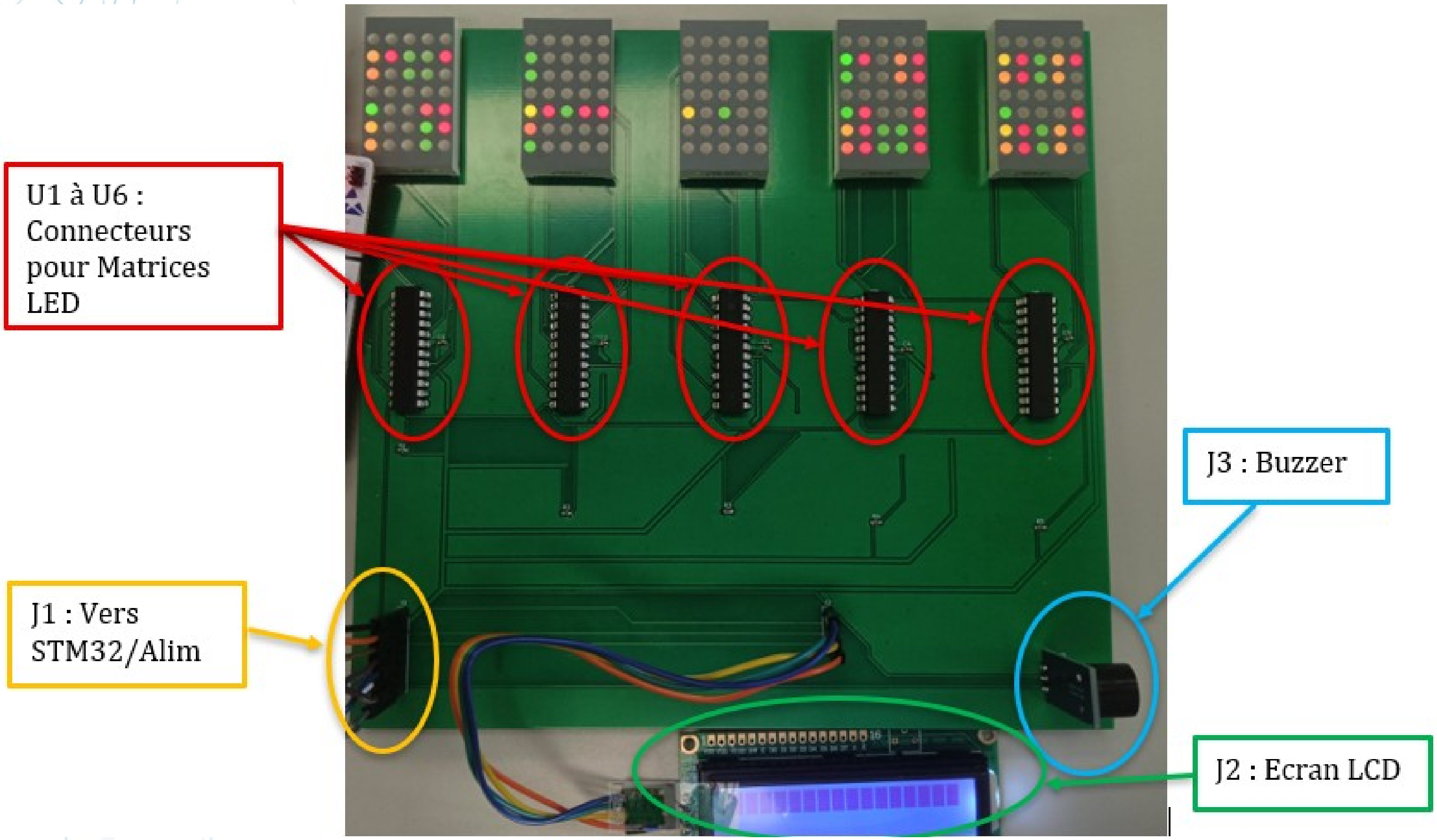
Figure 13 : PCB Acquisition assemblé



Short circuit identified after manufacturing

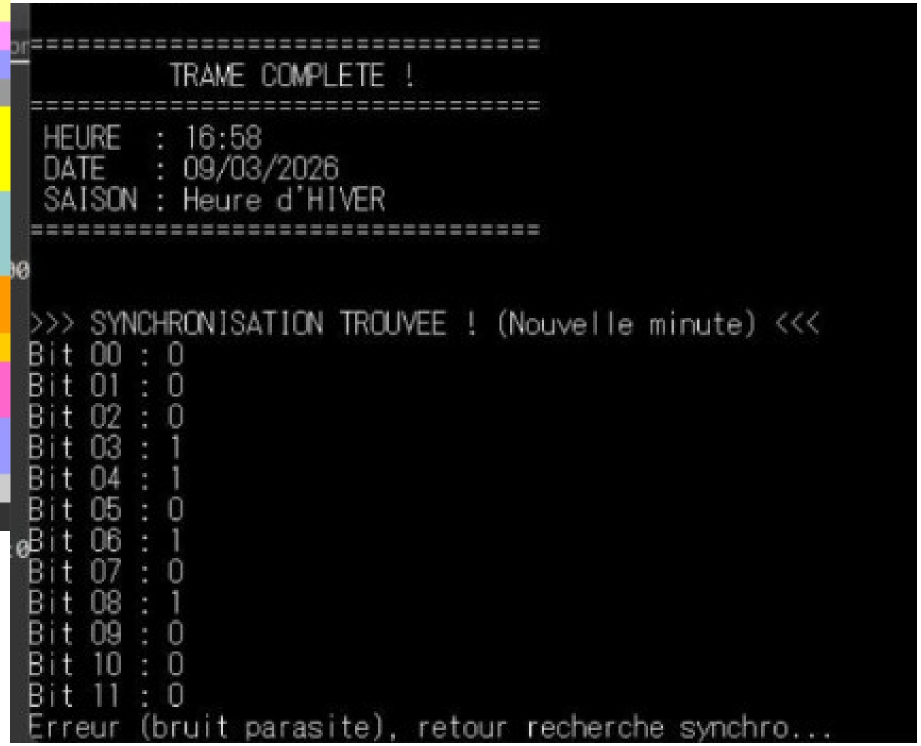
Impossible to make it work despite several attempts

DISPLAY


U1 à U6 :
Connecteurs
pour Matrices
LED
J1 : Vers
STM32/Alim
J3 : Buzzer
J2 : Ecran LCD

DISPLAY

Bit	Meaning
0	Starter bit
1..14	Weather info, special licence required
15	Special alert info for forces or technical assistance organisations
16	Marks change from standard to summer time
17..18	01: Standard time; 10: Summer
19	Switching second
20	Start of timing data
21..24	Minute data (unit position)
25..27	Minute data (decade)
28	Parity check bit (minute)
29..32	Hour data (unit position)
33..34	Hour data (decade)
35	Parity check bit (hour)
36..39	Day (unit position)
40..41	Day (decade)
42..44	Day of week (MON=0)
45-48	Month (unit position)
49	Month (decade)
50..53	Year (unit position)
54..57	Year (decade)
58	Parity check bit (Date), bits 36 to 57!
59	***ignored***



Successful time display on STM32CubeIDE

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LIFE CYCLE ANALISIS (SIMPLIFIED)

Elements	*Estimation*
Acquisition PCB	**20 à 40 g CO_2e**
Alimentation PCB	30 à 60 g CO_2e
Display PCB	250 à 600 g CO_2e
Acquisition components (SHT41, TEMT…)	20 à 80 g CO_2e
Alimentation components (L78, BU33…)	50 à 150 g CO_2e
Display components (LCD, LED matrix, RF antenna, buzzer…)	**500 g to 2 kg CO_2e**

Only extraction and manufacturing were considered !

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FEEDBACK

skills developed

Technical

→ Design a diagram that meets the need, choose the right footprints
→ Solder or glue the components onto the designated pads
→ define a global system, have a vision of the project

Management

→ manage a schedule, meet deadlines
→ Assign tasks, plan a backup solution in case of unforeseen events

Ecology/Technical

→ Raising awareness about the ecological transition by proposing viable technical solutions (Buck converter ? Only one PCB ? energy-saving mode ?)

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FEEDBACK

Areas for improvement for second-year projects

Management

→ We could have double-checked the KiCAD schematics before starting the manufacturing.

→ The start of manufacturing and assembly should have been done earlier to have at least two programming sessions

Ecology/Technical

→ We could have made a single PCB: this choice was not adopted because we wanted as many people as possible to be able to make a PCB

→ Reflection on a Buck converter (very little heat dissipation, better efficiency than a voltage regulator) but too expensive and lack of knowledge on this subject: Power Electronics Semester 7

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GREETINGS

Thank you for your attention, we are looking forward to your questions

Our Github's project: https://github.com/SidahmedHENNI/Horloge-Radio-Pilotee-DCF77

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